

FOR REFERENCE

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (EC) Sem. - III Examination, 2010-11
EC - 202 Network Theory

Date: 27/11/2010, Saturday

Time: 10:30 a.m to 01:30 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q-1(a) Define the following

4

i) Network ii) Node iii) Tree iv) Sub graph

(b) Explain Classification of Networks

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(c) What is the use of dot Convention? **Figure 1** shows three windings on a magnetic core. Using different shaped dots establish polarity markings for the windings.

4

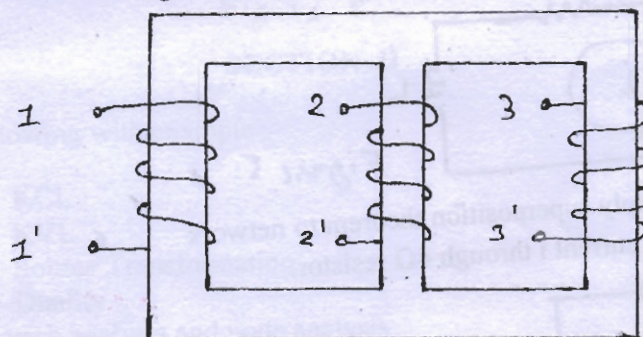


Figure 1

Q-2(a) In the network of **Figure 2**, the switch k is changed from position 1 to 2 at $t=0$. Find values of i , di/dt and d^2i/dt^2 at $t=0+$ if $R = 1000\Omega$, $L = 1H$, $C = 0.1\mu F$ and $V = 100v$.

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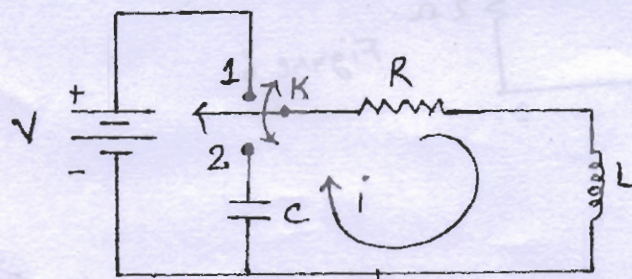


Figure:- 2

- (b) In the network of Figure 3, the switch k is closed at $t=0$. For the element values given obtain the general solution and particular solution for the current $i(t)$. Obtain the value of current at time $t=0.1$ sec.

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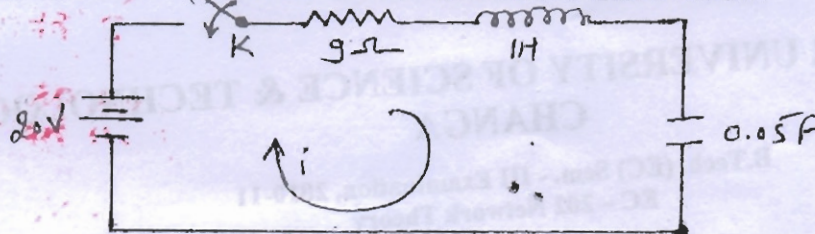


Figure:- 3

OR

- Q-2(a) In a network of Figure 4, a steady state is reached with the switch k open. At $t=0$, the switch is closed. For the element values given, determine the values of $v_a(0^-)$ and $v_a(0^+)$

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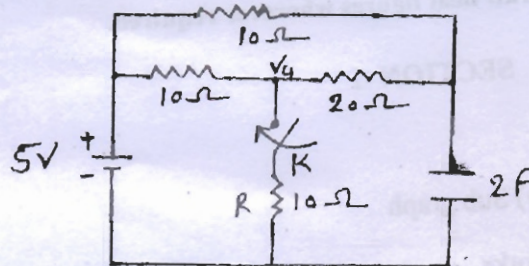


Figure 4

- (b) An exponential voltage $v(t) = 2e^{-4t}$ is applied at time $t=0$ to a series R-L circuit comprising a resistor $R=4\Omega$ and inductor $L=1H$ as shown in Figure 5. Obtain the particular solution for current $i(t)$ through the circuit. Assume zero initial current in the inductor

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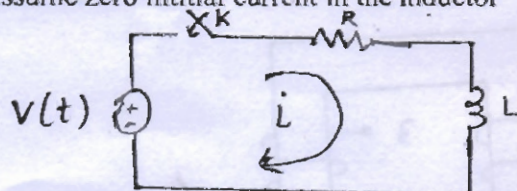


Figure 5

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- Q-3(a) State Superposition theorem. Apply superposition theorem to network shown in Figure 6 and find out current i through 4Ω resistor.

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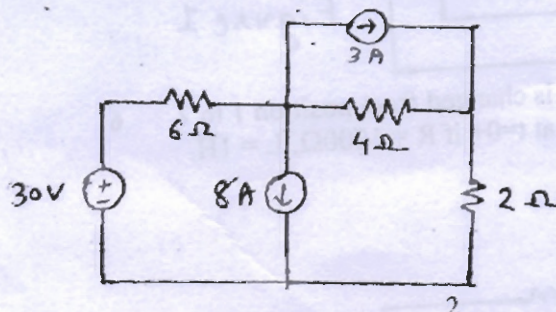


Figure 6

- (b) Define Poles and zeroes and find out poles and zeroes of the following function and locate it.

4

$$f(s) = \frac{s^2 + 2s + 1}{s^2 + 5s + 6}$$

- (c) How thevenin's theorem differs from Norton's theorem. Explain it.

2

OR

- Q-3(a) State Norton's theorem and find out the current i in the network of Figure 7 using Norton's theorem.

6

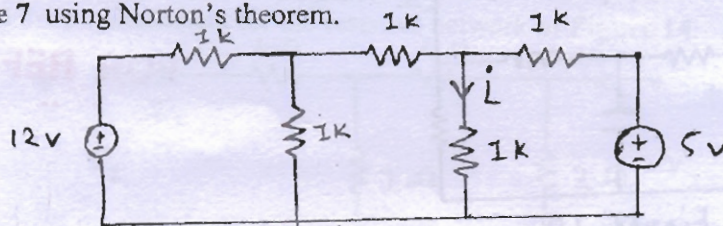


Figure 7

- (b) What do you mean by Maximum Power Transfer Theorem. Find out the condition for the Maximum Power Transfer Theorem

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- (c) Verify the reciprocity theorem for the network shown in Figure 8.

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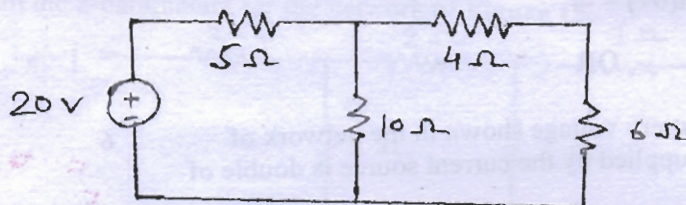


Figure 8

SECTION -II

- Q-4(a) Define following with example

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- i) KCL
- ii) KVL
- iii) Source Transformation
- iv) Duality

- (b) Compare mesh analysis and node analysis

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- Q-5(a) Use nodal analysis to find voltage across the 5Ω resistor in the Network of Figure 9.

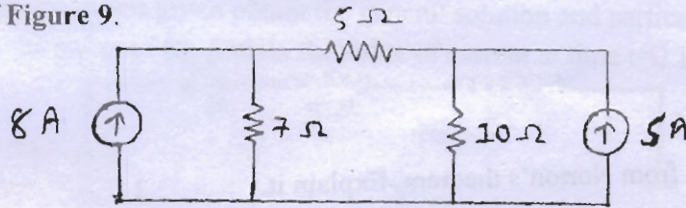


Figure 9

- (b) Draw the Dual of the Network shown in Figure 10

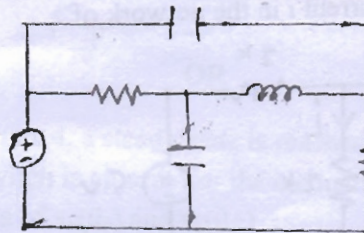


Figure 10

- (c) Using Laplace Transformation, solve the differential equation: 6

$$\frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + 8i = 8u(t)$$

Given that $i(0+) = 3$ and $di/dt(0+) = -4$

OR

- Q-5(a) Determine the value of the battery voltage shown in the network of Figure 11. when the power supplied by the current source is double of that supplied by the battery. 6

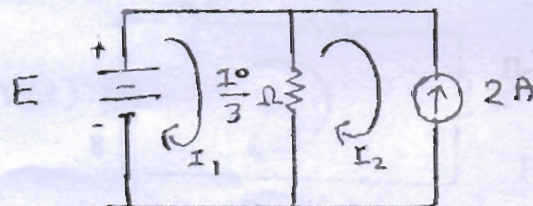


Figure 11

- (b) The waveform shown in Figure 12 is non recurring. Write an expression for this waveform and find the transform. 6

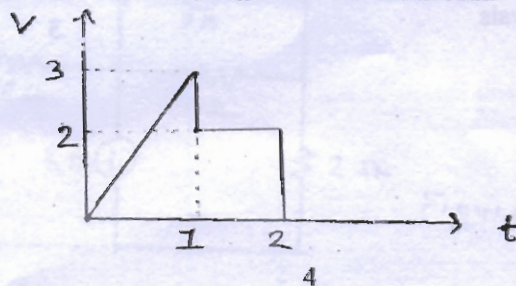


Figure 12.



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- Q-6(a) The waveform shown in **Figure 13** is a triangular waveform Determine the transform of this function. 6

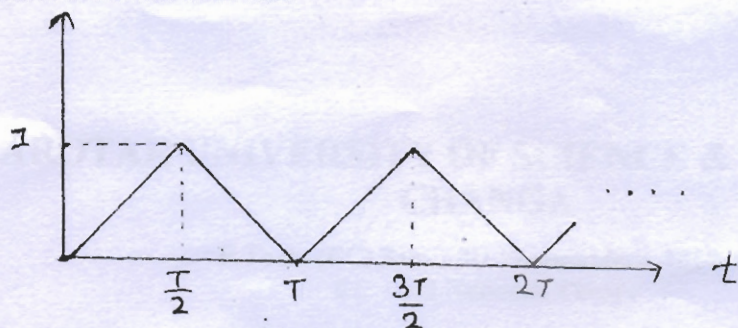


Figure 13

- (b) Find the y- parameters for the resistive network of **Figure 14**. 6

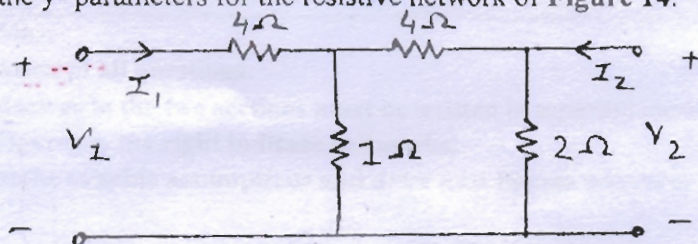


Figure 14

OR

- Q-6(a) Obtain the z-parameters for the network of **Figure 15**. 6

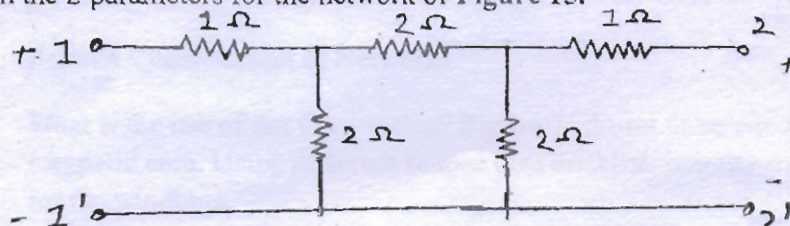


Figure 15

- (b) Explain relationship between y-parameters and ABCD parameters. 6